

Namit Shrivastava

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[LinkedIn](#) | [GitHub](#) | [Website](#)

RESEARCH SUMMARY

Graduate researcher with proven expertise in survey methodology, causal inference, and advanced statistical modeling. Conducted geospatial epidemiological research at the University of Michigan's Institute for Social Research by developing stratified analytical frameworks for large-scale census tract data with complex missing data patterns. Published survey research in peer-reviewed venues (Springer) and presented at AAPOR on transformer-based sentiment analysis methodologies. Academic background combines rigorous statistical training with practical survey research experience, achieving Dean's Fellowship recognition. Research agenda focuses on developing trustworthy data integration frameworks that fuse survey methodology principles with generative AI, deep learning architectures, and privacy-preserving computational methods to advance automated data collection, quality assurance, and responsible measurement at scale.

EDUCATION

University of Maryland, College Park

Master of Science, Survey & Data Science (Data Science Track)

Aug 2024 - May 2026

College Park, MD, USA

- **GPA: 3.817/4.0** | Dean's Fellowship Award Winner AY 2025-26
- Coursework: Statistical Modeling and Machine Learning (I and II), Fundamentals of Data Collection (I and II), Experimental Design and Causal Inference, Fundamentals of Computational Data Display, Long-Context Language Models, Machine Learning for Social Science, Applied Sampling, Applications of Statistical Modeling, Total Survey Error and Data Quality (I and II), Fundamentals of Inference, Analysis of Complex Sample Data, Practical Tools for Study Design and Inference, Survey Design Seminar, U.S. International Development and Foreign Aid Post-2025

Birla Institute of Technology and Science (BITS) Pilani

Bachelor of Engineering (Honours), Civil Engineering (Minor: Data Science)

Nov 2020 - Jul 2024

Pilani, RJ, India

- **GPA: 3.327/4.0**
- Coursework: Foundations Of Data Science, Object Oriented Programming, Operating Systems, Probability and Statistics, Applied Statistical Methods, Artificial Intelligence, Data Mining, Data Visualization, Human Resource Development, Technical Report Writing, Computer Programming, Principles of Management, Electrical Sciences, Machine Learning, Mathematics (I, II and III), Business Communication, Engineering Graphics, Fluid Mechanics, Construction Plan and Tech, Analysis of Structures

PROFESSIONAL EXPERIENCE

Institute for Mind and Brain, University of South Carolina

Data Scientist II

May 2026 - Present

Columbia, SC, USA

- Built a **HIPAA-compliant data platform** for an **NIH-funded longitudinal infant cohort**, using Python and R ETL pipelines to integrate REDCap API exports, ECG/IBI signals, temperature streams, behavioral coding, and clinical EHR data into **de-identified Parquet datasets** with automated

quality checks, schema validation, and audit logging.

- Engineered **multimodal time-series workflows** for ECG preprocessing, HRV/RSA extraction, behavior-physiology synchronization, and temperature biomarker computation, applying **MICE, mixed-effects models, latent growth curve analysis, and continuous-time Markov chains** for reproducible longitudinal biomarker discovery across **260 participants**.
- Designed and benchmarked **early risk stratification models** using scikit-learn pipelines with stratified cross-validation, RFECV, permutation importance, and ablation studies, alongside **PyTorch CNN-LSTM and Transformer models** evaluated with AUROC, AUPRC, bootstrap confidence intervals, and subgroup sensitivity analysis.
- Delivered an **AI-powered BI platform** using React, TypeScript, and Python with a grounded local LLM assistant to track cohort KPIs, enrollment trends, visit completion, attrition funnels, geospatial profiles, and ML model leaderboards, containerized with Docker and **deployed via Kubernetes, Helm, and Cloudflare Pages**.

Social Data Science Center, University of Maryland

Graduate Research Assistant

Jan 2026 - May 2026

College Park, MD, USA

- Engineered a Python and Playwright site-experience guard for the BSOS CKAN repository that crawled **592 same-origin links** across **420 HTML pages**, validated **7 public surfaces** in **2 responsive viewports**, and enforced screenshot, anchor, layout, and console checks to achieve a **100% pass rate** in the final live-site regression run.
- Built a GitHub Actions uptime and self-healing pipeline for a university data portal, monitoring **12 critical public routes** and **2 remote Docker containers** every **6 hours** with SSH-based repair logic and verifying **0 route failures** and **0 container failures** in production after deployment.
- Implemented Python-based data quality and release validation safeguards that audited **26 datasets** with **0 source-integrity issues**, verified deployment of **29 production override artifacts**, and backed the monitoring stack with **7 regression tests** plus **14 desktop/mobile visual baselines**.

Institute for Social Research, University of Michigan

Research Assistant

May 2025 - Dec 2025

Ann Arbor, MI, USA

- Architected and deployed a production-scale geospatial data integration pipeline processing **129,572 U.S. census tracts** across **3 RUCA** (Rural-Urban Commuting Area) strata, achieving **100% broadband data completeness** through multi-source fusion (FCC, ACS, CDC) and reducing baseline missingness by **28.6%** via composite measure imputation.
- Developed and validated a stratified epidemiological modeling framework analyzing COVID-19 incidence patterns across **2,788 census tracts** with **100% social determinant coverage**, uncovering statistically significant non-linear rural inflections ($p < 0.01$) and heterogeneous broadband effects by urbanicity.
- Designed and implemented a multi-stage data quality assurance protocol integrating Moran's I spatial autocorrelation analysis (identifying **15% spatial clustering violations**), temporal coverage validation (achieving **85% of tracts with $\geq 95\%$ completeness**), and RUCA-stratified missingness diagnostics.

Joint Program in Survey Methodology (JPSM)

Teaching and Graduate Assistant

Feb 2025 - May 2026

College Park, MD, USA

- Utilized Canvas LMS accessibility analytics through Blackboard Ally to identify, prioritize, and remediate high-severity digital barriers, ensuring **100% compliance** with **ADA and WCAG** standards across online course materials.

- Supported the launch of **2 brand-new 2026 courses**—**SURV 603: Benchmarking AI Models Using Surveys** and **SURV 605: AI-Assisted Surveys**—by managing Qualtrics configuration, Canvas module imports, syllabus preparation, accessibility reviews, and student welcome communications.
- Updated and maintained the **JPSM program website** and academic content infrastructure, ensuring accurate and timely distribution of **up-to-date syllabi** for upcoming semesters.
- Managed **Qualtrics-based registration pipelines** across multiple courses, strictly enforcing enrollment deadlines to maintain **accurate class rosters** prior to semester start.
- Liaised across program directors, course instructors, and IT scheduling teams to streamline Teaching Assistant onboarding and consistently meet a **48-hour SLA**.
- Assisted Dr. Jörg Drechsler from the Institute for Employment Research in teaching SURV735, guiding **23 students** in understanding data privacy and confidentiality principles.
- Optimized and coordinated learning operations using **Asana** and Canvas automation workflows, reengineering LMS infrastructure to reduce **course setup time by 40%** and improve semester launch readiness.

Legistify Services Private Limited

Machine Learning Engineer

Jan 2024 - Jun 2024

Gurugram, HR, India

- Engineered a scalable logo similarity detection system processing **2.4 million images** for intellectual property infringement analysis, implementing perceptual hashing (imagehash), feature extraction (SIFT), and approximate nearest neighbor search (Faiss) via FastAPI microservice to achieve **92% precision** and reduce manual review time by **70%**.
- Deployed a production Azure Cognitive Services API leveraging Vision Studio 4.0 for multi-modal document analysis (OCR, object detection, image captioning), processing **50,000+ legal filings** with **95% extraction accuracy** and enabling bilingual text recognition (English/Hindi) for **150+ client cases monthly**.
- Developed a phonetic trademark similarity algorithm using double metaphone and Jellyfish string matching to compare **50,000 new trademarks** against **300,000 existing database entries**, achieving **92% accuracy** in conflict detection while reducing computational overhead by **60%** through intelligent pre-filtering and optimized SQL queries.
- Integrated and stress-tested the Legistrak API ecosystem by executing **150+ REST API calls** via Postman with a **95% success rate**, identifying **12 critical bottlenecks** and implementing caching strategies that increased data retrieval efficiency by **30%** for **500+ daily active users**.

Accenture

Advanced Application Engineering Analyst

Jun 2023 - Aug 2023

Bangalore, KA, India

- Monitored enterprise security infrastructure by analyzing threat intelligence feeds from MITRE ATT&CK, AlienVault OTX, and proprietary sources, achieving **89% accuracy** in threat classification, enabling proactive defense against **25+ emerging vulnerabilities**, and contributing to a **20% reduction** in incident detection latency.
- Conducted penetration testing and vulnerability assessments on **50+ web applications**, networks, and cloud infrastructure using Burp Suite, Nmap, and Metasploit, identifying and documenting **120+ security vulnerabilities (15 critical, 45 high-severity)** that informed remediation roadmaps for Fortune 500 clients.
- Supported incident response operations during **3 security breaches** by performing log analysis with Kusto Query Language (KQL) and Azure Sentinel, assisting with containment and forensics, contributing to post-incident reports, and reducing mean time to recovery (MTTR) by **80%** through automated runbook implementation.
- Developed Python automation scripts for log parsing, threat indicator extraction, and security metrics

dashboarding, processing **10M+ daily log events** and generating executive-level threat summary reports that enabled data-driven security decision-making for C-suite stakeholders.

Indian Red Cross Society

Web Developer

May 2022 - Jul 2022

Bangalore, KA, India

- Designed and deployed a Drupal-based content management system for volunteer registry and donor database management, streamlining volunteer onboarding workflows and reducing manual data entry time by **50%**.
- Localized the IRCS main website by implementing multilingual support for Hindi and Kannada regional languages, conducting functionality, performance, and security testing, and achieving a **95% bug-free** user experience across **10,000+ monthly visitors**.

PUBLICATIONS AND PROJECTS

Achieving Sustainability in Supply Chain during Disruption Times: Role of Industry 4.0

Advances in Data-Driven Computing and Intelligent Systems (Springer)

Aug 2023 - Feb 2024

BITS Goa, India | [\[Publication Link\]](#)

- Designed and executed a qualitative survey study with **200+ automotive supply chain experts** across India, employing fuzzy-set Analytical Hierarchy Process (FAHP) to rank **5 Industry 4.0 technologies** and **6 green supply chain practices**. Achieved consistency ratios below **0.10 threshold**, validating expert judgment reliability across all matrices.
- Developed a hierarchical Interpretive Structural Modeling (ISM) framework integrating MICMAC analysis to map causal relationships between digital technologies (IoT, Big Data Analytics, Cloud Computing) and sustainable practices. Identified government regulation and top management commitment as key drivers with highest driving power (score of **9 out of 9**).
- Established quantitative linkage between Industry 4.0 adoption and green supply chain performance through three-level hierarchical structure. Demonstrated that IoT, BDA, and cloud computing function as critical linkage factors (**driving power 7, dependence power 7**), enabling effective supplier-customer collaboration and circular economy practices.

Analyzing Public Sentiments of EVs in the Era of Sustainable Transformation

80th AAPOR Annual Conference

Nov 2024 - May 2025

St. Louis, MO | [\[Conference Paper\]](#)

- Engineered a multi-source sentiment analysis pipeline processing **1.1M+ social media posts** from **5 Reddit communities (550K comments)** and **40 New York Times articles**. Implemented DistilBERT transformer model achieving **91.6% classification accuracy** on electric vehicle discourse spanning 2020-2024.
- Discovered systematic bias in large language model sentiment prediction through comparative analysis of **10 Groq LLM variants** (Llama 3.1/3.2 series, Mixtral). Found LLMs exhibited **+0.57 positive sentiment bias** compared to actual Reddit data (**M=-0.18, SD=0.52**), with statistical significance (**F(2,549)=28.43**).
- Identified temporal trends showing **35% increase in negative EV sentiment** on Reddit post-2022 despite rising adoption rates, while NYT coverage remained consistently negative (**M=-0.23**) for core EV terms but positive for Tesla and autonomous keywords. Demonstrated that Llama-3.2-90b-vision-preview best approximated human sentiment patterns.

Player Market Value Analysis in Elite European Football

Graduate Research Project

Sep 2025 – Dec 2025

University of Maryland, College Park, MD, USA | [\[LinkedIn\]](#)

- Developed a cross-classified multilevel model analyzing **7,023** player-season observations from **2,159** elite European football players, explaining **78%** player-level variance and showing squad-status playing time drives a **42%** market valuation premium over traditional performance metrics.
- Engineered a cubic polynomial time-series model capturing non-linear market dynamics (**23.7** quadratic contraction; **11** cubic correction), isolating COVID-19's structural impact on the **€38bn** transfer market and quantifying valuation gaps (**65%** goalkeeper discount; **34%** East Asian discount; **21%** Latin American premium; **17%** domestic-citizen discount).

English Premier League (1992–93 to 2021–22): Spending Analysis and 20-Year Forecasting

Course Project – Sports Analytics

Jan 2025 – Apr 2025

University of Maryland, College Park, MD, USA | [\[LinkedIn\]](#)

- Automated extraction, cleaning, and integration of **30** EPL seasons (1992–2021) into a unified dataset with **600+** club-season records to support advanced analytics and long-horizon forecasting.
- Built and evaluated Random Forest, ARIMA, Prophet, LSTM, and SVR models, achieving **72.3%** out-of-sample classification accuracy and generating **20-year** forecasts; quantified that higher average annual expenditure corresponds to finishing **2–4** places higher and visualized a **500%** increase in average club expenditure since league inception.

Air Pollution Abatement and Modeling

Undergraduate Thesis Project

Aug 2023 – Dec 2023

Birla Institute of Technology and Science (BITS) Pilani, Pilani, Rajasthan, India | [\[LinkedIn\]](#)

- Conducted an air quality monitoring study in Pilani using APM 460/APM 550 cyclone-based respirable dust samplers, processing **26** ambient air samples across **766** hours and estimating PM₁₀ mean **184.2** µg/m³ with peaks **307.6** µg/m³; observed exceedance of national standards by **64.5%**.
- Developed ML forecasting models (Linear Regression, Decision Tree, Random Forest) and implemented a Gaussian plume dispersion model achieving **R²=0.87** with RMSE **12.4** µg/m³ for PM₁₀ (and **R²=0.85**, RMSE **8.7** µg/m³ for PM_{2.5}), evaluating **4** emission-reduction scenarios with up to **24%** projected pollution reduction.

Applications of ML/IoT in Concrete Technology and Structural Health Monitoring

Undergraduate Research Project

May 2023 – Dec 2023

Birla Institute of Technology and Science (BITS) Pilani, Pilani, Rajasthan, India | [\[LinkedIn\]](#)

- Developed a multilabel CNN using ResNet50/ResNet101 on **42,520** road-surface images (1920×1080) across **5** distress categories, achieving **86.67%** accuracy, **88.19%** F1-score (80/20 split) and reducing manual road inspection time by an estimated **90%**.
- Expanded training data from **52,009** to **245,735** labeled distress instances via augmentation and deployed a multilabel classifier using binary cross-entropy with sigmoid activation, supporting multi-distress images and enabling scalable road monitoring workflows.

LEXNet: Lightweight Explainable CNNs for Internet Traffic Classification

Undergraduate Research Project

Aug 2023 – Dec 2023

Birla Institute of Technology and Science (BITS) Pilani, Pilani, Rajasthan, India | [\[LinkedIn\]](#)

- Architected LEXNet for real-time traffic classification across **1.5M** network flows and **200** application classes, matching ResNet-level accuracy while reducing model size by **97%** (to **118,880** parameters) and cutting inference time by **93%** for deployment on resource-constrained routers.
- Engineered LERes and LProto components reducing backbone parameters by up to **19%–36%** and CPU inference time by up to **24%–41%**, while improving accuracy by **4%** and providing faithful prototype-based explanations that outperform ProtoPNet and LSTM/ResNet baselines.

Madhav Gadgil vs. Kasturirangan Reports: Western Ghats Conservation Policy Analysis

Course Project

Apr 2023

Birla Institute of Technology and Science (BITS) Pilani, Pilani, Rajasthan, India | [\[LinkedIn\]](#)

- Conducted environmental policy analysis comparing two Western Ghats conservation frameworks across **6** states, **44** districts, and **142** taluks; quantified ESA coverage differences (Gadgil: **75%** / **137,000** hectares vs. Kasturirangan: **37%** / **60,000** hectares), a **50.7%** reduction in protected coverage.
- Analyzed governance and zoning approaches (Gadgil ESA-1/2/3 allocation: **60%/15%/25%** vs. Kasturirangan emphasis on **90%** natural landscape protection with **8%** cultural-use allocation), highlighting a shift from bottom-up Gram Sabha governance to centralized working-group execution.

Active Learning for Forest Cover Classification

Course Project – Machine Learning

Sep 2022 – Dec 2022

Birla Institute of Technology and Science (BITS) Pilani, Pilani, Rajasthan, India | [\[LinkedIn\]](#)

- Implemented an active learning framework on a forest cover dataset of **581,012** observations with **54** features, achieving **85%** accuracy using Entropy Sampling and outperforming random sampling by approximately **15** percentage points for **7** cover types.
- Compared **5** active learning strategies and built a Query-by-Committee ensemble (Random Forest + Decision Trees), demonstrating uncertainty-based sampling reduces labeling needs by **60–70%** while maintaining **82–85%** model accuracy.

Automated Plagiarism Detector for Text and Images

Course Project – Artificial Intelligence

Sep 2022 – Dec 2022

Birla Institute of Technology and Science (BITS) Pilani, Pilani, Rajasthan, India | [\[LinkedIn\]](#)

- Developed an automated plagiarism detection system implementing the Winnowing algorithm with a dual-search architecture (local database + MSN Live Search), supporting essays, code, and podcast transcripts; achieved **68.1%** accuracy with a LinearSVC classifier and enabled template fingerprint removal to reduce false positives.
- Engineered an image plagiarism module using Average Hash (A-Hash) with 8×8 grayscale hashing and automated Excel reporting with hash-difference visualization, reducing manual plagiarism review time by an estimated **95%**.

Statistical Analysis and Forecasting of Solar Energy

Course Project – Applied Statistical Methods

Sep 2022 – Dec 2022

Birla Institute of Technology and Science (BITS) Pilani, Pilani, Rajasthan, India | [\[LinkedIn\]](#)

- Engineered a solar energy forecasting system on a **14-year** dataset (2000–2014) with **8** meteorological parameters, identifying correlations exceeding **0.80** for key GHI factors and selecting beta distributions via Kolmogorov–Smirnov testing.
- Built AR/MA/ARMA/ARIMA/SARIMA models for solar radiation prediction, achieving **5.93%** MAPE using SARIMA on **781** weekly observations and demonstrating strong accuracy with lower computational cost than ML baselines.

Voice Gender Recognition using Acoustic Features

Independent Data Science Project

Jul 2022

Birla Institute of Technology and Science (BITS) Pilani, Pilani, Rajasthan, India | [\[LinkedIn\]](#)

- Developed a gender recognition model on **3,168** voice samples with **20** acoustic features, benchmarking **5** ML algorithms and achieving **98.3%** classification accuracy using Random Forest (50 estimators).
- Tuned a Random Forest classifier (entropy criterion; max_depth=6; n_estimators=50) achieving **98.6%** training and **98.3%** testing accuracy on a balanced dataset (**1,584** male / **1,584** female),

with only **11** misclassifications.

TECHNICAL SKILLS

Programming: Python, R, Java, C++, JavaScript, TypeScript, HTML/CSS, Bash/Shell

Data & Databases: SQL (MySQL, PostgreSQL), NoSQL (MongoDB, Cassandra), Snowflake, Neo4j, Pinecone, Apache Spark, Kafka

AI/ML: PyTorch, TensorFlow, Keras, Hugging Face, LangChain, LlamaIndex, LangGraph, PySpark, NLTK, Spacy, SkLearn

DevOps: Git, Docker, Kubernetes, Jenkins CI/CD, REST APIs, Django, Flask, Angular, React.js, Terraform, Ansible

Cloud: AWS (EKS, S3, EC2, Lambda, Sagemaker), Azure, GCP, Tableau, Power BI, MLFlow, Qualtrics, STATA

Core: LLMs, Generative AI, RLHF/DPO, Deep Learning, NLP, Computer Vision, Survey Methodology, Causal Inference, MLOps

LEADERSHIP & SERVICE

Terrapin Leadership Institute, University of Maryland

Aug 2024 - May 2025

Member

College Park, MD

- Achieved **100% participation** in workshops on leadership, ethics, inclusion, and resilience.
- Enhanced group dialogue quality by contributing reflective insights in all sessions, fostering more inclusive and engaging discussions among peers.

National Service Scheme (NSS), BITS Pilani

Mar 2022 - Dec 2023

Executive Committee & Blood Donation Camp Core Team Member

Pilani, RJ, India

- Planned **10+ activities** to improve English skills and confidence in villages around campus.
- Organized blood donation camp coordinating **60+ volunteers**, managing data for **1,000+ donors**, achieving **844 successful donations** within 2 days.

Peer Mentorship Program (PMP), BITS Pilani

Aug 2021 - Dec 2023

Mentor

Pilani, RJ, India

- Assisted Juniors with advice and materials to ensure they have a smooth transition into college life and succeed academically.

CERTIFICATIONS & ACHIEVEMENTS

Certifications: Microsoft Certified: Azure AI Fundamentals | API-based Programming (Postman) | AI For Everyone (Coursera) | Deep Learning & Machine Learning (Smartknower, Internshala)

Achievements: **1st rank** in Human Resource Development (**180 students**, BITS Pilani) | **Top 10** in Water & Wastewater Treatment (**98 students**, BITS Pilani) | JPSM Dean's Fellowship Award (AY 2025-26, UMD)